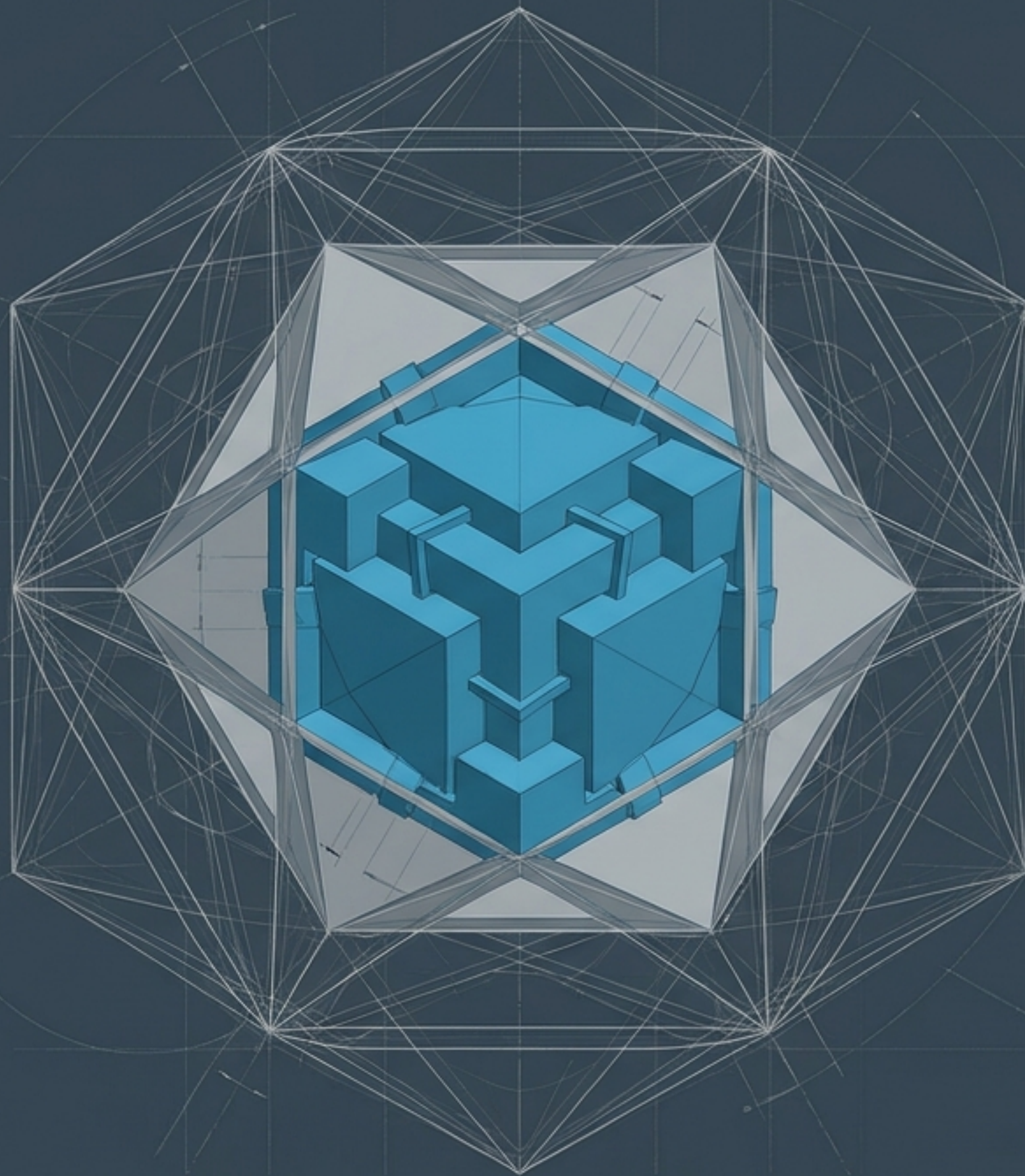


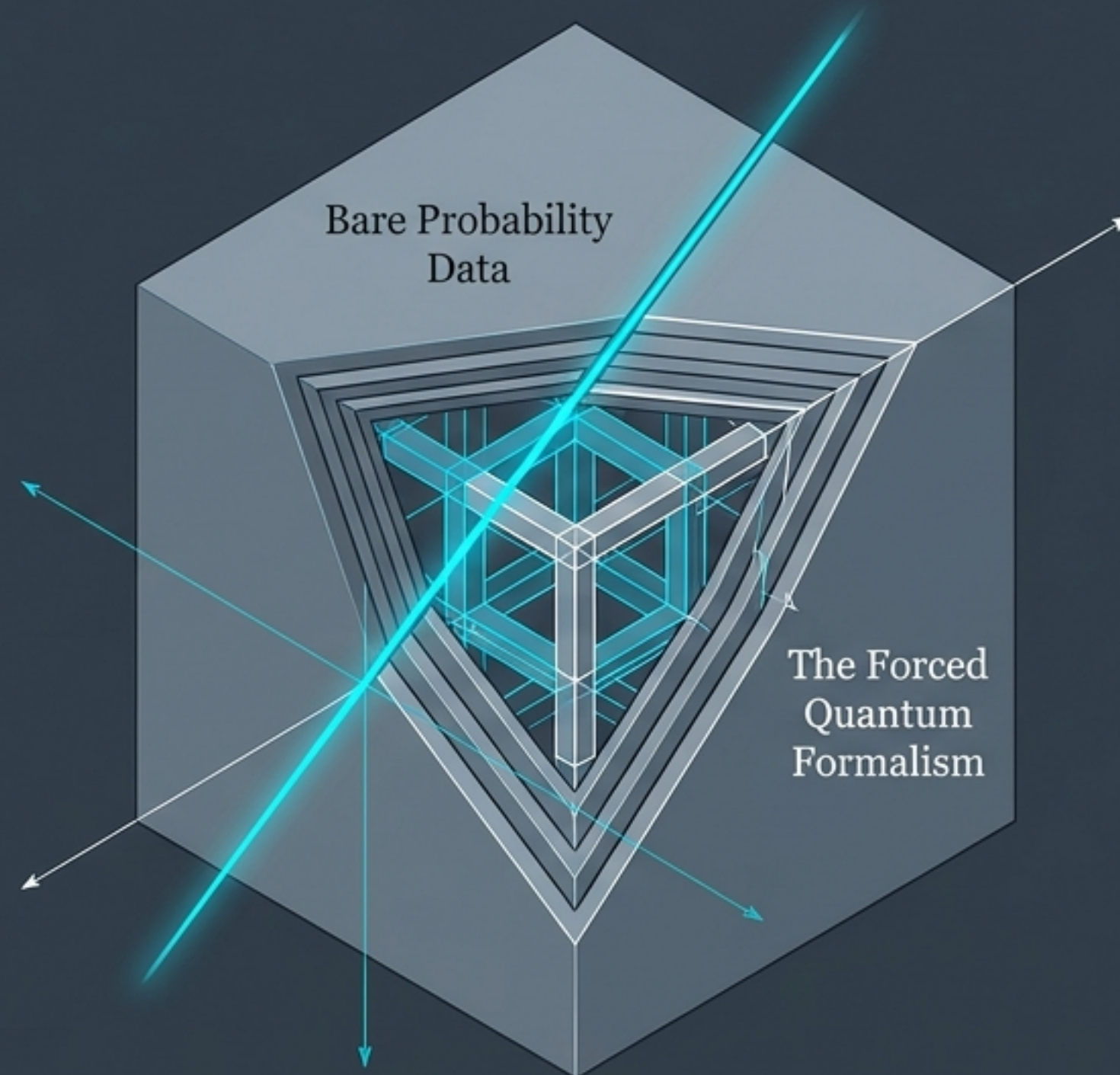
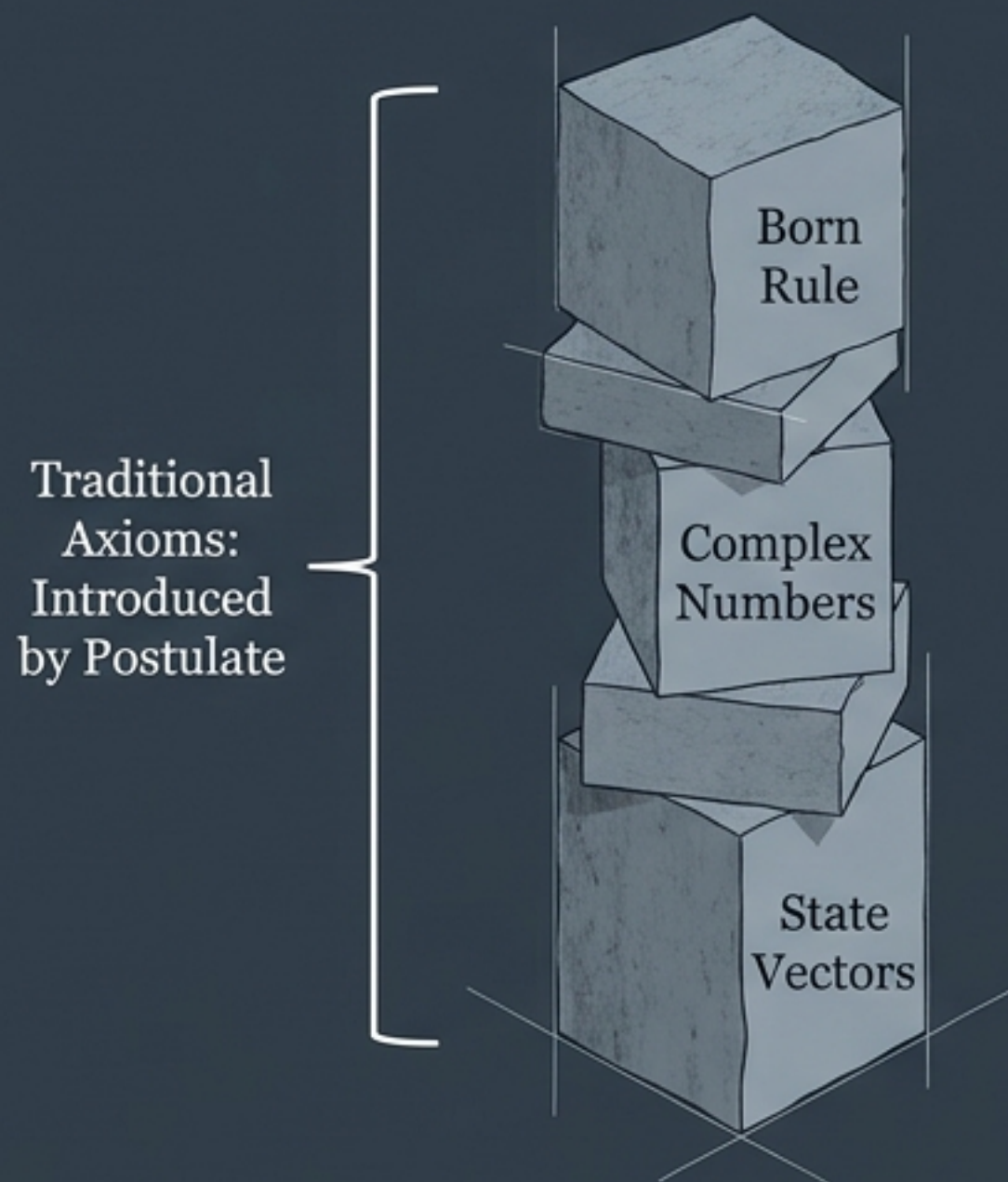


The Architecture of Constraints

Recovering Quantum
Mechanics from Admissible
Probabilistic Composition

[D] Derived | [A] Assumed | [U] Unresolved

Reconstruction over Recovery



We ask a different question:
If we start only with the admissibility constraints on probabilistic data, how
much of the quantum formalism becomes strictly necessary?

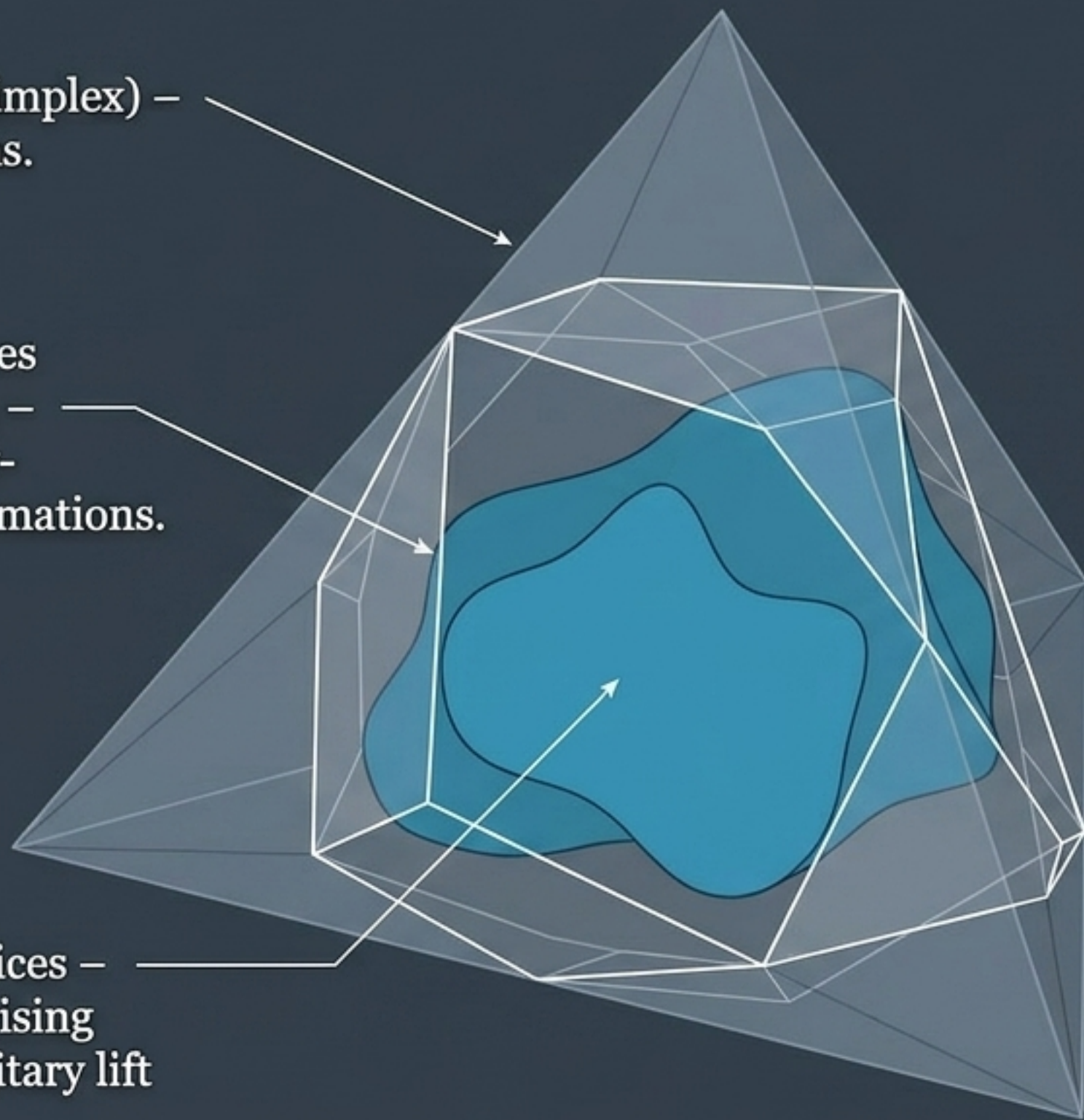
[A] The Probability Simplex
(Starting point prior to
amplitude or phase)

The Geometry of Admissible Transitions

Stochastic Maps (Simplex) –
Arbitrary transitions.

Bistochastic Matrices
(Birkhoff Polytope) –
Reversible, entropy-
preserving transformations.

Unistochastic Matrices –
Transformations arising
from a coherent unitary lift
($B_{ij} = |U_{ij}|^2$).

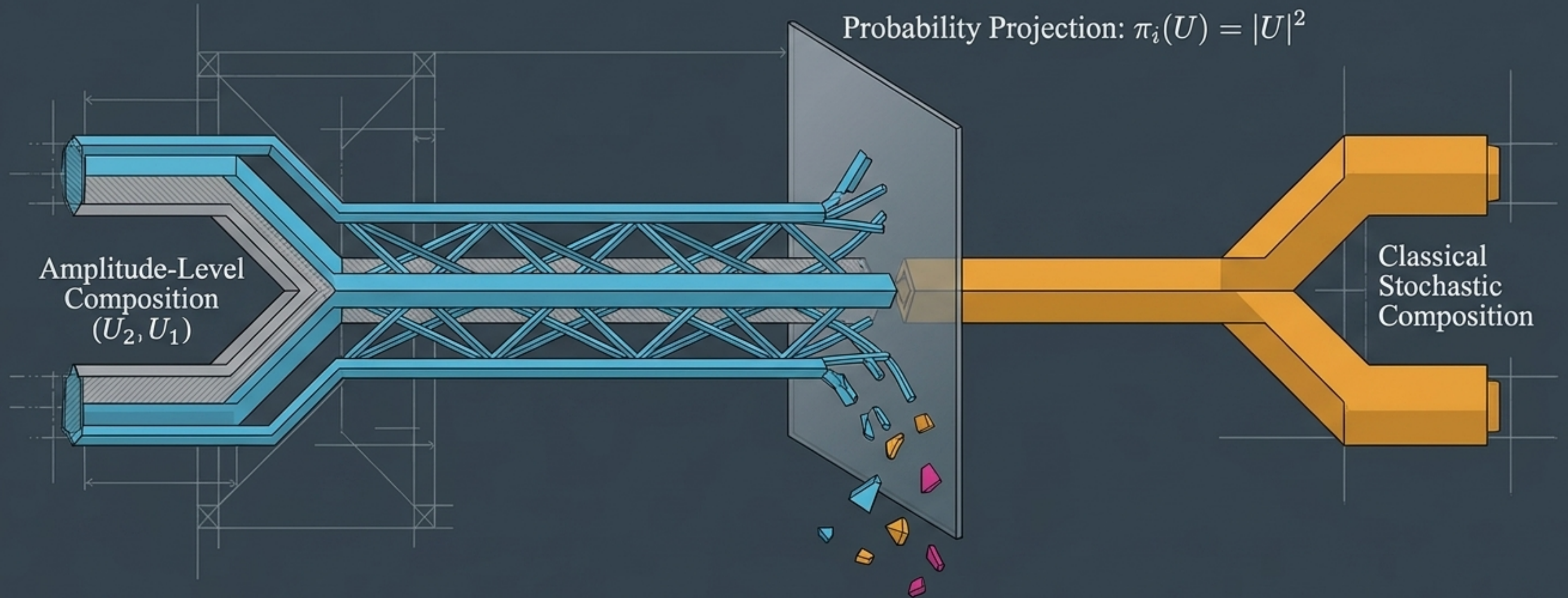


The classical restriction to bistochastic maps yields a polytope generated by permutations. But at $n \geq 3$, the unistochastic subset requires continuous, non-convex geometry.

The 3×3 Discrete Fourier Transform matrix explicitly admits no real lift, separating complex/real at the single-matrix level.

[D] Bistochasticity as Majorization
[D] $n=3$ Real Incompleteness

Axis 1: The Composition Obstruction

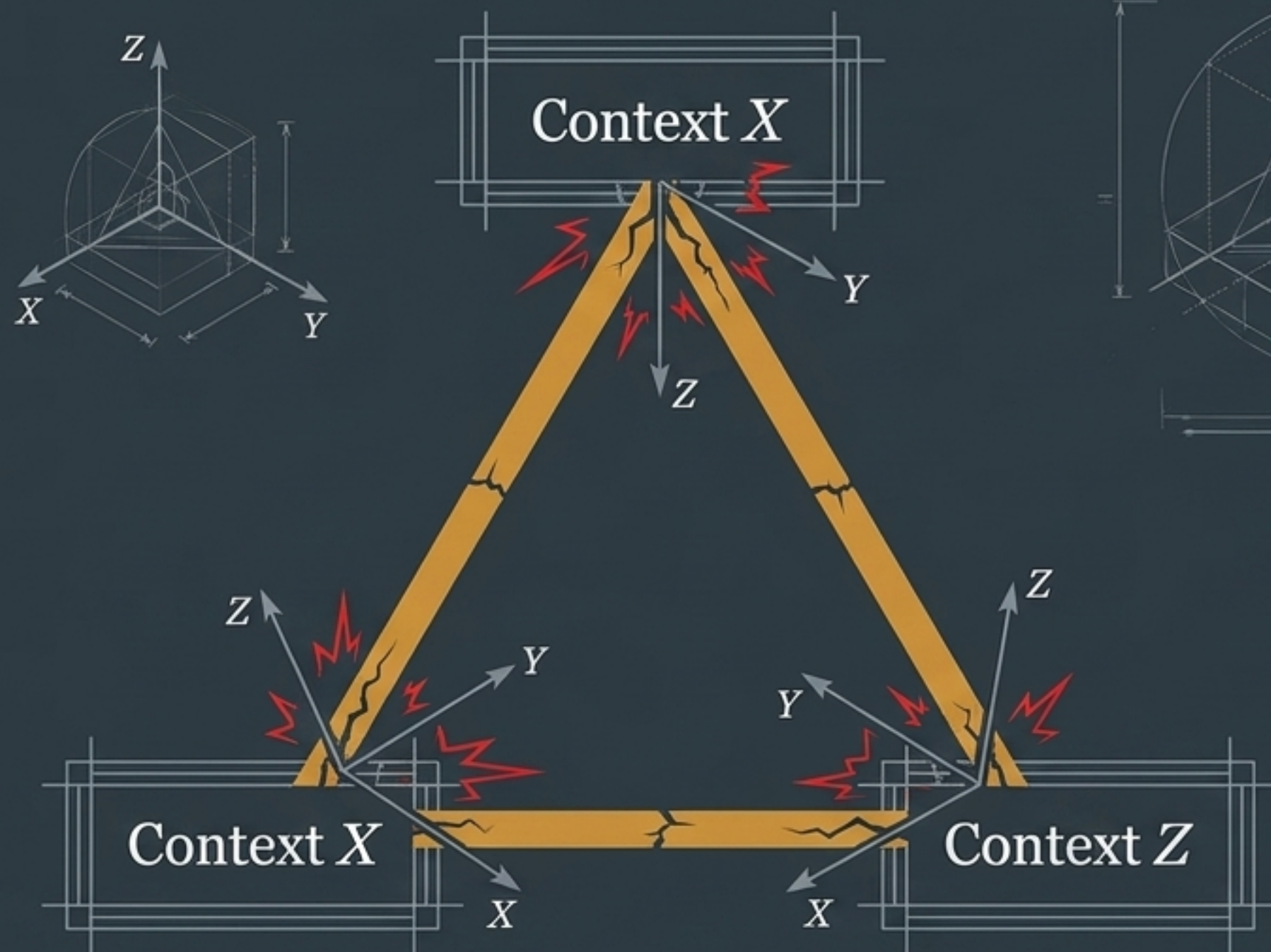


Composing transformations at the probability level discards cross-term information. A coherent lift restores this compositional closure. However, interference alone does not select Complex over Real.

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[D] Path Loss under p_i
[U] Complex Necessity via Composition Alone

Axis 2: Contextual Cycle Gluing



Edgewise real admissibility does not guarantee global real admissibility.

The smallest graph-theoretic witness to this failure is the K_3 triangle.

Three individually admissible transition edges cannot glue into a single real realization due to cycle-rank constraints.

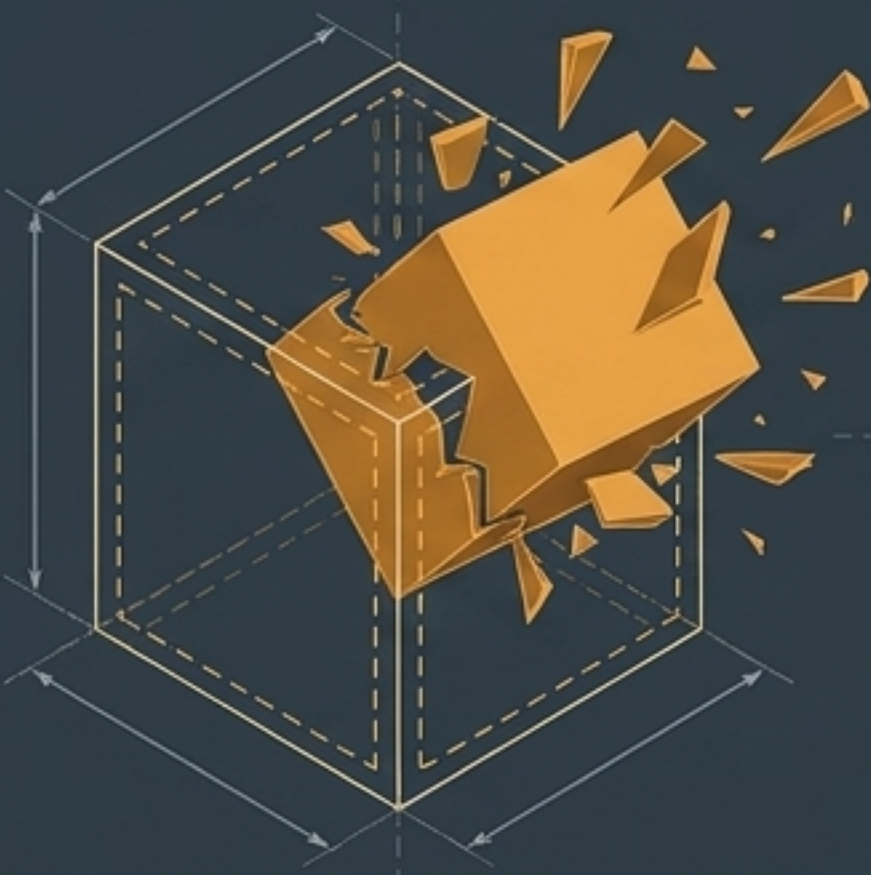
The Mutually Unbiased Bases (MUB) in $\mathbb{R}^2$ fail to align globally.

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[D] Triangle-Gluing Theorem
[D] Real MUB Obstruction

Axis 3: Local Tomography and the Dimension Mismatch

How must independently accessible systems compose? Only over Complex fields is a joint state exactly as informative as complete local measurements on its subsystems.

Real (R)	Quaternionic (H)	Complex (C)
		
Mismatch = $\frac{d_A d_B (d_A - 1)(d_B - 1)}{4} > 0$	Mismatch = $-2 d_A d_B (d_A - 1)(d_B - 1) < 0$	Mismatch = 0

Overshoots: Invisible holistic parameters.

Undershoots: Overdetermined product space.

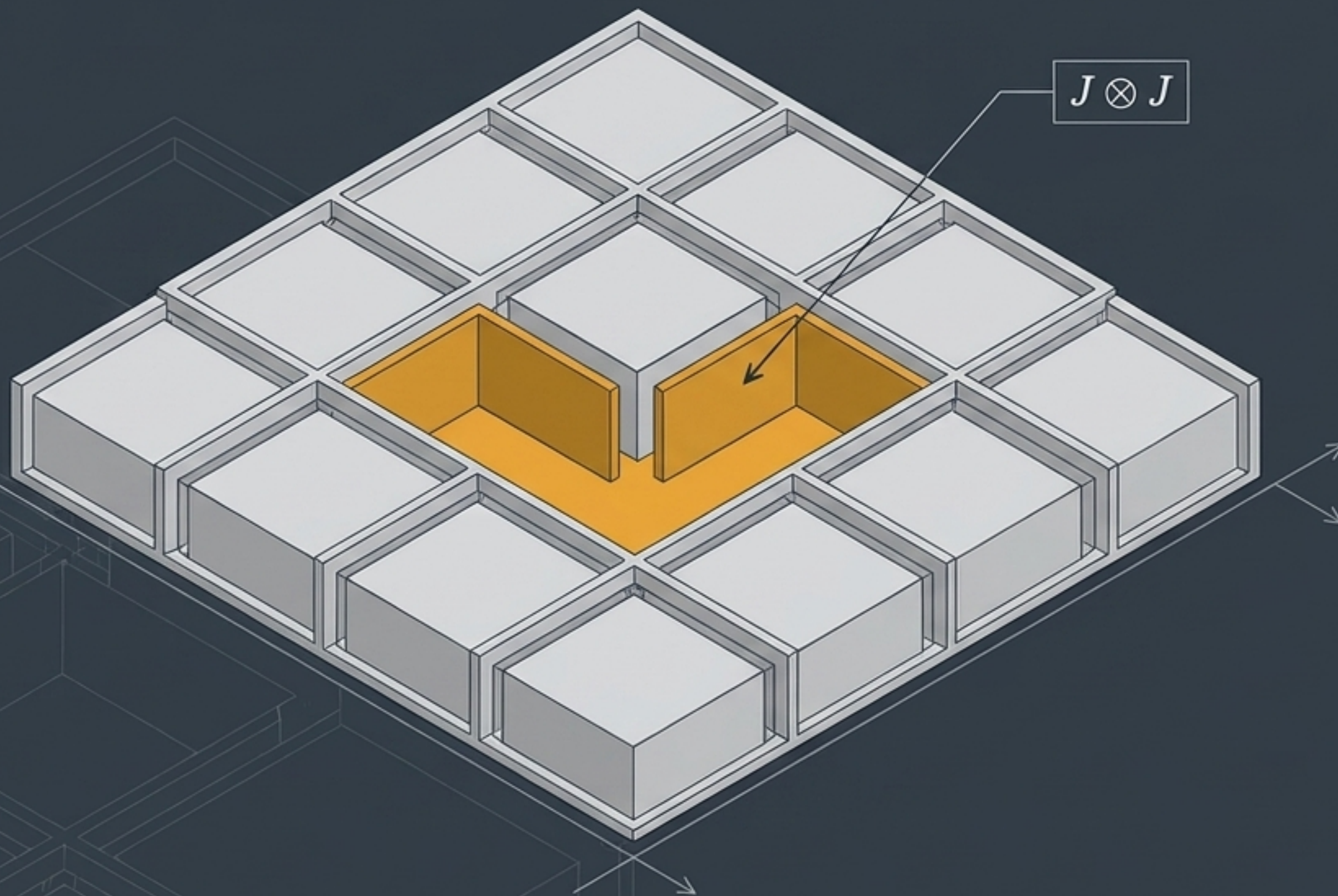
The Goldilocks Match: Exact dimensional alignment.

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[A] Local Tomography Principle
[D] Dimension-counting identity selects C

NotebookLM

The Invisible Real Dimension

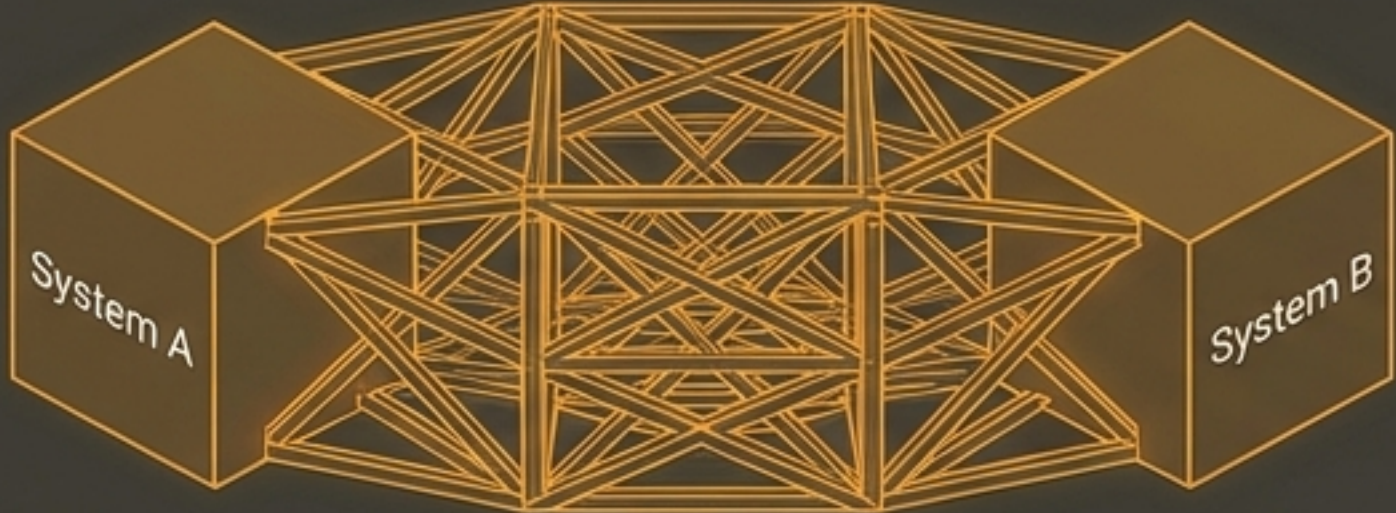
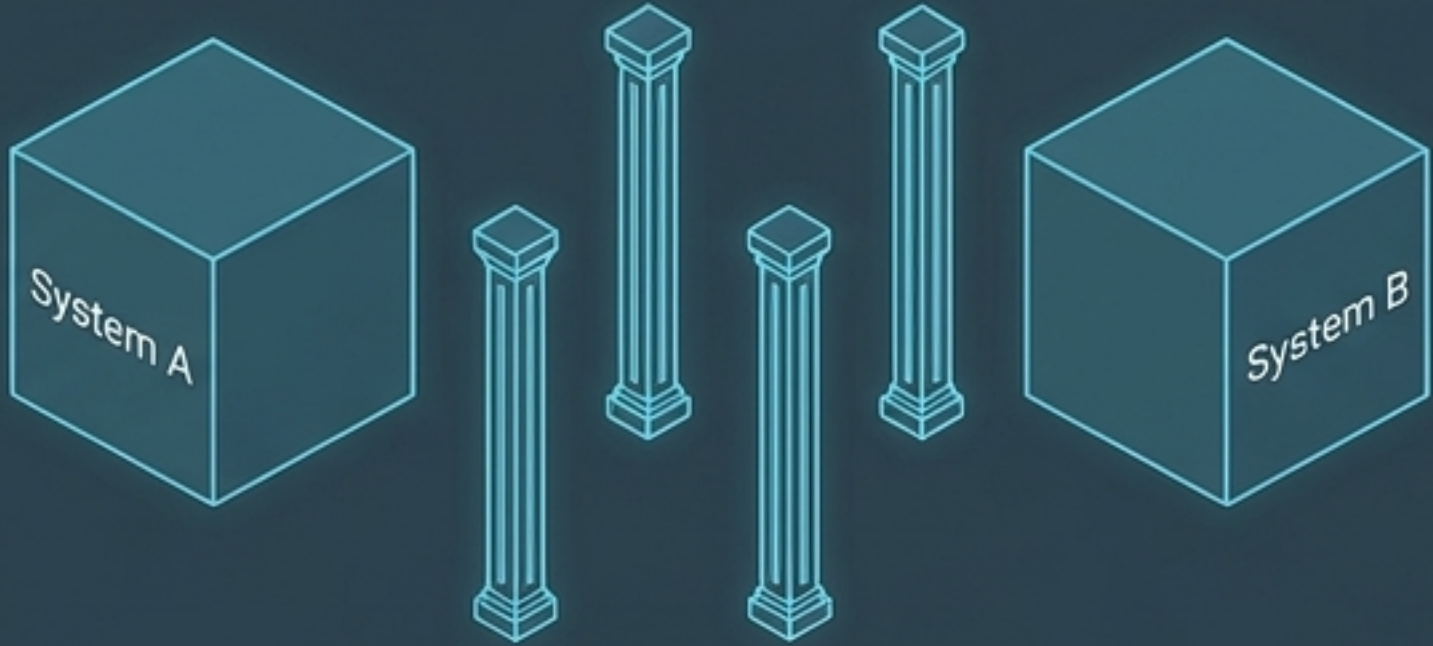


In $\mathbb{R}$, the missing dimension of composite space is spanned by $J \otimes J$ (where J is the antisymmetric generator). It exists in the ambient composite space but is invisible to any measurement built from local real observables.

Operational positivity bounds this locally invisible parameter to a physical one-parameter family of states: $\rho(c)$ where $|c| \leq 1$.

[D] $J \otimes J$ basis element
[D] Positivity Bound

Entanglement is Field-Relative

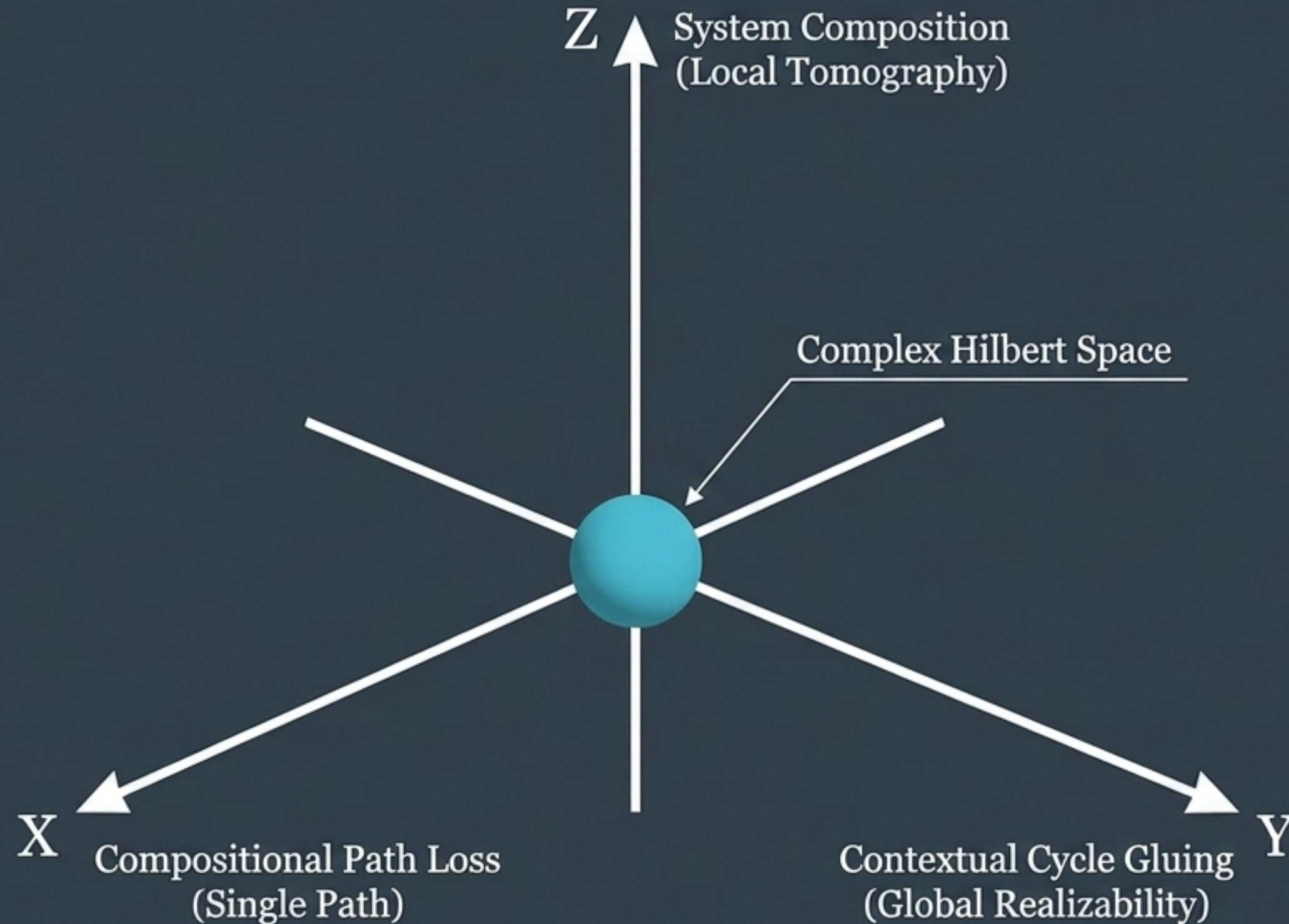
Evaluated under $\mathbb{R}$	Evaluated under $\mathbb{C}$
 $\text{Tr}[\rho(J \otimes J)] = c$	
R-Entangled (for all $c \neq 0$)	C-Separable (for all c in $[-1,1]$)

The exact same physical family of states is entangled when composed over $\mathbb{R}$ (via the $J \otimes J$ witness), yet entirely separable when composed over $\mathbb{C}$. Both partial traces annihilate this coefficient—proving that global physical reality does not imply local recoverability.

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- [D] Partial Trace Annihilation
- [D] Field-Relative Separability

The Unique Intersection



Complex Quantum Mechanics
isn't just one solution to one
problem.

It is the unique intersection that
simultaneously resolves three
mathematically independent
geometric and operational
failures of classical probability.

[R] Synthesis of Parts I & II

The Born Hierarchy

The Born rule is not a single axiom. It is a hierarchy of four distinct statements. The pure-state vector rule is simply one column of the context-transition identity, costing exactly one operational assumption.

$$B_{ij} = |U_{ij}|^2$$

Supported by:
Unistochastic
Admissibility

$$p_i = |\langle e_i | \psi \rangle|^2$$

Supported by:
Pure States as
frame-completable
vectors

$$p_i = \text{tr}(\rho P_i)$$

Supported by:
Affine mixtures

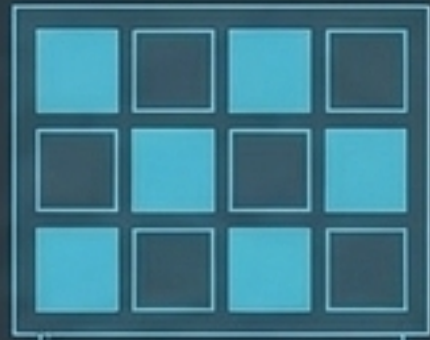
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- [A] Pure states as frame-completable vectors
- [D] Density-operator Born rule

Observables as Packaged Projectors

The Operational Bridge

Mutually orthogonal
projectors
(Sharp Measurement, P_a)
 $P_a P_b = \delta_{ab} P_a$



1, 3,
2, 7,
...
 $m, 10$

Real number
labels (a)

$$A = \sum a P_a$$

Spectral
Theorem

Hermitian
Operator (A)

"Observables are Hermitian" is not an independent postulate. It is an operational equivalence.

Assigning real-valued labels to sharp projective measurements inherently constructs a Hermitian operator. The Eigenvalue rule immediately follows as a corollary.

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[D] Hermitian/Sharp Measurement Equivalence
[A] Real-valued outcome labels

Time and the Generator Theorem

$$U(t+s) = U(t)U(s) \rightarrow K^\dagger + K = 0 \rightarrow U(t) = e^{-itH/\hbar}$$



Once we assume time evolution is continuous and reversible, it must take the form of a strongly continuous one-parameter unitary group. Schrödinger's equation is merely the infinitesimal form of this continuous composition law.

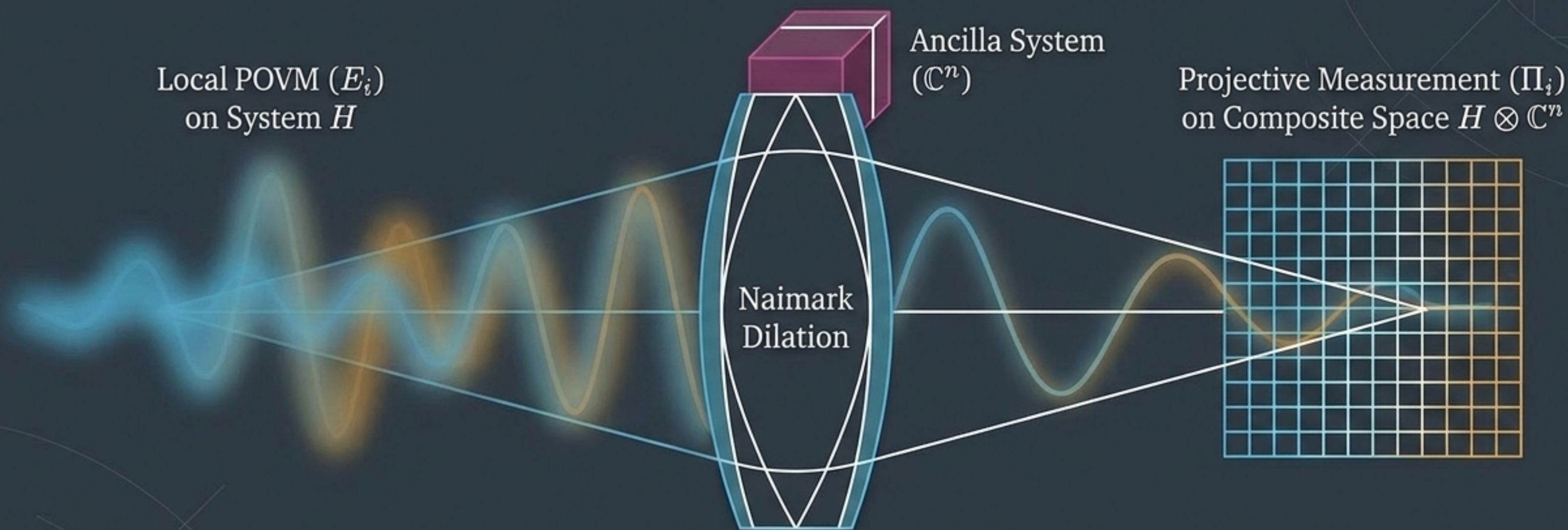
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[A] Continuous reversible time evolution

[D] Schrödinger's Equation

[U] H = Energy

Generalized Measurement as Local Perspective



Every generalized measurement (POVM) is simply a sharp projective measurement performed elsewhere. It utilizes the exact same system-plus-ancilla architecture as the partial trace, operating in the complementary direction.

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[D] Naimark dilation for finite POVMs

The Immutable Ledger

Beyond the bare mathematical constraints of admissible probabilistic data, the full recovery of finite-dimensional quantum theory cost exactly four operational assumptions:



1. Local Tomography (tested, not mathematically forced)
2. Pure states are frame-completable vectors
3. Measurement outcomes are labeled by real numbers
4. Time evolution is a strongly continuous unitary group

Ledger UI stamp

[D] The entirety of the
finite-dimensional formalism.
All other postulates are strictly derived.